When should you rebalance?

Private capstone research | 7 November 2026 | No default change

Annual ranks first by utility-adjusted net wealth among the four schedules tested. But there is no risk-compliant multi-year optimum established. At three and five years, no frequency passes all four validation laws under the original loss budgets. A higher score is not enough to call it the best way to invest.

One year: annual means hold within the year

Your confirmed amount is $250k/month. Twelve modeled calls total $3M. On fresh final paths, every frequency passes the common risk/funding screen under all four laws. The target starts with 76.39% cash, so the whole year of payments needs no sales under hold. This is not a general annual-rebalancing recommendation.

Final net CARA CE, $M	Monthly	Quarterly	Semiannual	Annual / hold
IID nominal	98.217	98.224	98.230	98.246
Bounded nominal	98.209	98.216	98.224	98.241
Drift -2pp/year	97.946	97.950	97.953	97.962
Drift -2pp / cov x1.2	97.949	97.953	97.956	97.964

Annual minus monthly: roughly $29k IID, $32k bounded, $16k/$15k in lower-drift laws. Only the bounded-law comparisons are clearly separated by the conservative simultaneous 12-comparison Monte Carlo intervals. Elsewhere intervals can include zero. These are simulation intervals, not economic confidence.

What is fixed

$100M research balance sheet: 4.581% US equity, approximately zero EM/gold, 9.033% Treasuries, 76.385% cash and 10% unsold Singapore property. Cash/rent are zero. Four fitted synthetic return laws, not historical out-of-sample returns. Long-only, no borrowing or property sale. These are research inputs, not a fully calibrated personal plan.

5bps on each buy/sell; one-week sale settlement; buys only from settled cash. Payments reduce cash once; pending receipts cannot pay early bills. These costs/timing are modeled assumptions, not verified broker rules. Annual still funds monthly payments. CE means terminal NET CARA certainty equivalent with $10M tolerance, not mean wealth or return.

Multi-year score is not a pass

Three and five years were tested after your yes to a multi-year comparison. They are duration sensitivities, not newly confirmed obligations: 36/60 modeled $250k calls total $9M/$15M. Annual now rebalances between years. No loss budget was loosened and no target was refitted.

3-year final comparison

Net CARA CE, $M	Month	Quarter	Half-year	Annual
IID nominal	94.457	94.478	94.501	94.552
Bounded nominal	94.470	94.489	94.516	94.572
Drift -2pp/year	93.660	93.673	93.685	93.713
Drift -2pp / cov x1.2	93.668	93.681	93.693	93.720

Validation selection: NONE. Only bounded nominal final rows pass. All four frequencies fail the path-loss screen under the other three laws.

5-year final comparison

Net CARA CE, $M	Month	Quarter	Half-year	Annual
IID nominal	90.705	90.735	90.771	90.866
Bounded nominal	90.655	90.689	90.734	90.829
Drift -2pp/year	89.403	89.420	89.438	89.494
Drift -2pp / cov x1.2	89.421	89.438	89.454	89.510

Validation selection: NONE. All 16 frequency/law final rows fail the path-loss screen.

All payments are funded in every displayed final generated path. The failure is investment-path risk, not a lack of cash for bills. Annual leads by CE across every tested law: about $54k-$102k over three years and $89k-$174k over five years versus monthly. This is a ranking of tested schedules, not a feasible optimum.

Interpretation

A longer horizon changes how much investment risk accumulates. Rebalancing timing alone does not rescue the original target under these fixed loss budgets. Long-run stationarity, unchanged horizon-unit EP views and constant cost/settlement assumptions are unvalidated extrapolations. Do not compare terminal CEs across horizons as annualized returns: withdrawals and time at risk differ.

Fresh validation 4x200 and final 4x400 paths per law, per horizon. Disjoint seeds and selection locks; no final repair. Paired four-batch intervals use a simultaneous 24-comparison correction and can be wide. Nominal IID modeled fees: annual $1,486/$2,948 versus monthly $5,301/$8,745 over three/five years. The study took 14.72 seconds.

Did dynamic feedback win?

No proven best dynamic policy. We searched a causal eight-parameter policy instead of imposing a months-of-cash playbook. Its primary nominal gain mostly came from changing the initial allocation, not valuable later feedback.

Primary search and post-hoc attribution

Final common net CE, $M	IID nominal	Bounded nominal
Original cash-first	98.252660	98.240332
Learned repeated feedback	98.810853	98.832610
Same first trade, then cash-first	98.839217	98.852218

All these rows pass the original two nominal final screens. But applying the same first trade then stopping learned feedback beats repeated feedback by about $28k IID and $20k bounded. This is post-hoc attribution on the same original paths, not an independent confirmatory test. The learned policy fails the prespecified -2pp drift path-risk check.

Exploratory four-law robust search: no candidate

After that failure, a frozen robust redesign searched 360 parameter vectors against the four fixed laws. Forty evaluated candidates passed training screens. NONE of the top 12 passed all four validation laws. No robust policy was selected; no extra search was added after final results. This is failure of a bounded search, not proof that no suitable policy exists.

Fresh final all-risk check	IID	Bounded	-2pp	-2pp / cov1.2
Original cash-first	PASS	PASS	PASS	PASS
Pro-rata withdrawals	PASS	PASS	PASS	PASS
Static monthly target	PASS	PASS	PASS	PASS
Original feedback	PASS	PASS	FAIL	FAIL
Original first trade only	PASS	PASS	FAIL	FAIL

On fresh lower-drift paths, BOTH original feedback and first-trade-only fail path-loss. The older attribution pass for first-trade-only under one drift sample did not hold up. All these rows still fund every payment: paid-all alone is not the full risk test.

No dynamic global optimum or near-optimality certificate

A clairvoyant frictionless weekly relaxation bounds shared-sample terminal wealth from above. Its sample CE is about $241M IID and $197M bounded, far above roughly $98M policy scores. It is unattainable and very loose: there is no useful near-optimality certificate. This is not an achieved return or a confidence interval on the true-law optimum.

The math and its boundary

The problem is finite-horizon stochastic control, with causal actions and a joint funding constraint. An exact solution is conditional on the specified model; simulation cannot establish what is universally best.

State and admissible actions

State includes liquid units, settled cash, pending sale receipts, fixed property units, arrears, remaining known calls and observed market history. A fully risk-constrained dynamic program also needs running path-loss/payment-event statistics. Sales cannot exceed holdings. Purchases plus fees cannot exceed settled cash. Sales enter the settlement queue; due calls debit cash; arrears persist.

Objective: maximize -T log E[exp(-X_H/T)], with T=$10M and X_H = cash + marked remaining units/property + pending receipts - arrears. Paid withdrawals are already deducted. Controls cannot observe future returns or random future liabilities. The implementation rejects scenario-varying future liability amounts.

Constraints and actual search

Long-only, no leverage, fixed unsold property; joint paid-all target 99%; dated arrears and shadow path/terminal CVaR-type screens under reference and entropy-pooled nominal/stressed measures. The same executed-policy judge is applied to every baseline. An old additive W1 reserve is a heuristic stress screen here, NOT a valid nonlinear-feedback DRO certificate. No confidence-calibrated uncertainty ball is claimed.

Four risky logits relative to cash, plus shared coefficients for funding ratio, remaining fixed liabilities, observed drawdown and trailing four-week momentum. Eight bounded parameters; differential evolution, 40-vector population, eight generations, 360 evaluations, no polishing. Training penalty is a search device, not an exact constrained solver. Validation screens the top 12 training-feasible candidates. Primary train100 / validation4x200 / final4x400 paths per law.

What was actually checked

Accounting conservation with fees and pending receipts; no future-price leakage; delayed cash cannot pay early; same first action for attribution; pro-rata withdrawal arithmetic; strict judge rejects unpaid calls; random future liabilities/rent rejected; actual multi-year annual turnover; fixed snapshot truth; pathwise relaxation dominance. A tiny two-period action-grid Bellman calculation equals exhaustive nonanticipative enumeration. That equality proves only that tiny finite problem, not the full policy optimum.

Remaining work before economic conclusions

A tighter certified bound or exact solvable formulation; realistic calibrated costs, taxes and settlement; a justified long-run return/expense law; chronological out-of-sample evaluation. Property index returns and assumed dependence are not executable property liquidity. No client preference parameter was estimated. No default, holding or account was changed.

Evidence and sources

The source code, frozen protocols, all search trials, validation failures, selection locks and final summaries are preserved in the private repository. The companion shows a frozen research panel for one/three/five-year comparisons. Edited dashboard controls do not recompute this dynamic study.

Verification: 368 tests passed with three existing numerical warnings. Built and installed the wheel outside the source checkout; checked packaged research/data and the executed annual ledger. Desktop and 320px phone layouts, horizon/law switches and failure labels were inspected; no JavaScript errors. All five PDF pages were visually checked.

Reproduce

```
python -m experiments.dynamic_execution
python -m experiments.dynamic_attribution
python -m experiments.dynamic_robust
python -m experiments.dynamic_bounds
python -m experiments.dynamic_frequency
python -m experiments.dynamic_multiyear
python -m experiments.dynamic_science_bundle
```

The initial primary sensitivity run had a source-constructor error. Its failed selection lock and technical retry note are preserved. Corrected rerun exactly matched all original 360 trials and validation selections. No final-data tuning. Later input hardening and wider simultaneous frequency intervals change no point estimates or selected frequency.

Research sources

Skaf/Boyd (2009): constrained multi-period control with transaction costs

Lobo/Fazel/Boyd (2007): linear/fixed transaction costs and relaxations

Boyd et al. (2014): performance bounds and suboptimal policies

Dupacova (1999): output analysis and misspecification

Dupacova/Consigli/Wallace (2000): multistage scenarios

Kuchler/Vigerske: out-of-sample evaluation of approximations

These primary academic sources support formulation and evaluation methods. Their quadratic/convex special-case theorems do NOT prove optimality for this CARA, settlement and joint-chance-constrained policy class. Model parameters and simulation outcomes come from the preserved experiment, not from those papers.

Conditional evidence, not a guarantee

Each displayed final row has 1600 independent reference paths per frozen law. With every path funded, the one-sided 95% binomial lower bound is 99.8129%. That bound concerns generated paths under the frozen fitted model, not actual investment outcomes. Weighted EP rates are descriptive, not binomial trials. Multiple laws are sensitivities, not an estimated probability distribution over future regimes.